Dynamic Multi-Value Parameter Passing in Oracle BIP via OIC — From For-Each to XSLT Optimization

Context : In the context of an OIC integration calling an Oracle BI Publisher report, the integration reads a flat file containing multiple headers, each referencing a PO Number. The BIP report is used to retrieve purchasing data for enrichment purposes.

Objective : Call a single BIP report with all PO Numbers extracted from the input file, in order to retrieve purchasing data for all headers in one single API call.

Problem Statement : The initial approach used a For Each loop to iterate over each header and call the BIP report individually — one call per PO Number. This approach works but raises a performance concern: for a file containing N headers, the integration makes N BIP calls.

The challenge is therefore :

- How to **collect all PO Numbers dynamically** from the parsed input file
- How to **pass them as a single parameter** to the BIP report
- How to make the **BIP SQL query** handle a comma-separated list of values instead of a single value
- How to **eliminate the For Each** loop and replace it with a smarter XSLT-based approach

1. Input File Structure :

1.1. File Format

The input file is a semicolon-delimited flat file encoded in UTF-8.

Each field is surrounded by double quotes. The file contains two types of lines identified by the first field value:

HDR : Header line containing receipt-level data including the PO Number
LIN : Detail line containing item-level data linked to its header via RECEIPT_ID

Sample File : testInput.txt


```
##########################################

"HDR";"S0063";"PIE";"VoltEdge";"MAGASIN";"10070";"HI1";"CL";"98";"";
"202605220700";"S";"202604031617";"44";"";"202603201617";"";"";"
```

";" ";""
"LIN";"REC";"HI1";"23";"S109";"S109";"Value";"Value";"2";"U";"UNIT";"1
";"1";"1";"S";"";"";"";"";"";"";"";"1";"";"1";"0";"999999";"0";"0";"";" "
"HDR";"S0063";"PIE";"VoltEdge";"MAGASIN";"10070";"HI2";"CL";"226";""
;"202605220700";"S";"202604031617";"44";"";"202603201617";"";"";"
";" ";""
"LIN";"REC";"HI2";"18";"S109";"S109";"Value";"Value";"2";"U";"LIT";"1";
"1";"1";"S";"";"";"";"";"";"";"";"1";"";"1";"0";"999999";"0";"0";"";" "

##

1.2. NXSD Schema

The flat file is parsed in OIC using a Native Format Schema (NXSD). The schema defines an xsd:choice element with nxsd:choiceCondition="terminated" and nxsd:terminatedBy=";" to route each line to the correct element based on the first field value : - nxsd:conditionValue="HDR" → parsed as Header element - nxsd:conditionValue="LIN" → parsed as Line element All fields are terminated by ";" with quoteChar="""" The last field of each line is terminated by ${eol}

##

```
<?xml version = '1.0' encoding = 'UTF-8'?>

<xsd:schema xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:nxsd="http://xmlns.oracle.com/pcbpel/nxsd"

        xmlns:tns="http://TargetNamespace.com/ServiceName"
targetNamespace="http://TargetNamespace.com/ServiceName"

        elementFormDefault="qualified"
attributeFormDefault="unqualified" nxsd:version="NXSD"
nxsd:stream="chars"

        nxsd:parseBom="true"

        nxsd:encoding="UTF8">

  <xsd:element name="RCV">

    <xsd:complexType>

      <xsd:choice minOccurs="1" maxOccurs="unbounded"

              nxsd:choiceCondition="terminated"
```

```
                nxsd:terminatedBy=";">
        <xsd:element name="Header" nxsd:conditionValue="HDR">
          <xsd:complexType>
            <xsd:sequence>
              <xsd:element name="WAREHOUSE_CODE"
type="xsd:string"
                        nxsd:style="terminated"
                        nxsd:terminatedBy=";"
                        nxsd:quoteChar="&quot;"/>
              <xsd:element name="ACTIVITY_ZONE_CODE"
type="xsd:string"
                        nxsd:style="terminated"
                        nxsd:terminatedBy=";"
                        nxsd:quoteChar="&quot;"/>
              <xsd:element name="THIRD_PARTY_CODE"
type="xsd:string"
                        nxsd:style="terminated"
                        nxsd:terminatedBy=";"
                        nxsd:quoteChar="&quot;"/>
              <xsd:element name="ACTIVITY_CODE" type="xsd:string"
                        nxsd:style="terminated"
                        nxsd:terminatedBy=";"/>
              <xsd:element name="SUPPLIER_CODE" type="xsd:string"
                        nxsd:style="terminated"
                        nxsd:terminatedBy=";"
                        nxsd:quoteChar="&quot;"/>
              <xsd:element name="RECEIPT_ID" type="xsd:string"
```

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                nxsd:style="terminated"
                nxsd:terminatedBy=";"/>
        <xsd:element name="RECEIPT_TYPE" type="xsd:string"
                nxsd:style="terminated"
                nxsd:terminatedBy=";"
                nxsd:quoteChar="&quot;"/>
        <xsd:element name="PO_NUMBER" type="xsd:string"
                nxsd:style="terminated"
                nxsd:terminatedBy=";"
                nxsd:quoteChar="&quot;"/>
        <xsd:element name="REFERENCE_2" type="xsd:string"
                nxsd:style="terminated"
                nxsd:terminatedBy=";"
                nxsd:quoteChar="&quot;"/>
        <xsd:element name="EXPECTED_RECEIPT_DATE"
type="xsd:string"
                nxsd:style="terminated"
                nxsd:terminatedBy=";"
                nxsd:quoteChar="&quot;"/>
        <xsd:element name="STATUS" type="xsd:string"
                nxsd:style="terminated"
                nxsd:terminatedBy=";"
                nxsd:quoteChar="&quot;"/>
        <xsd:element name="ACTUAL_RECEIPT_DATE"
type="xsd:string"
                nxsd:style="terminated"
                nxsd:terminatedBy=";"
```

```
                nxsd:quoteChar="&quot;"/>
        <xsd:element name="EDI_ID" type="xsd:string"
                nxsd:style="terminated"
                nxsd:terminatedBy=";"
                nxsd:quoteChar="&quot;"/>
        <xsd:element name="TRANSPORT_HEADER"
type="xsd:string"
                nxsd:style="terminated"
                nxsd:terminatedBy=";"
                nxsd:quoteChar="&quot;"/>
        <xsd:element name="ARRIVAL_DATE" type="xsd:string"
                nxsd:style="terminated"
                nxsd:terminatedBy=";"
                nxsd:quoteChar="&quot;"/>
        <xsd:element name="SHORT_LABEL" type="xsd:string"
                nxsd:style="terminated"
                nxsd:terminatedBy=";"
                nxsd:quoteChar="&quot;"/>
        <xsd:element name="REFERENCE_3" type="xsd:string"
                nxsd:style="terminated"
                nxsd:terminatedBy=";"
                nxsd:quoteChar="&quot;"/>
        <xsd:element name="REGISTRATION_CODE"
type="xsd:string"
                nxsd:style="terminated"
                nxsd:terminatedBy=";"
                nxsd:quoteChar="&quot;"/>
```

```
            <xsd:element name="SEAL_CODE" type="xsd:string"
                        nxsd:style="terminated"
                        nxsd:terminatedBy=";"
                        nxsd:quoteChar="&quot;"/>
            <xsd:element name="BUILDING_CODE" type="xsd:string"
                        nxsd:style="terminated"
                        nxsd:terminatedBy="${eol}"
                        nxsd:quoteChar="&quot;"/>
          </xsd:sequence>
        </xsd:complexType>
      </xsd:element>
      <xsd:element name="Line" nxsd:conditionValue="LIN">
        <xsd:complexType>
          <xsd:sequence>
            <xsd:element name="PROCESSING_TYPE" type="xsd:string"
                        nxsd:style="terminated"
                        nxsd:terminatedBy=";"
                        nxsd:quoteChar="&quot;"/>
            <xsd:element name="RECEIPT_ID" type="xsd:string"
                        nxsd:style="terminated"
                        nxsd:terminatedBy=";"
                        nxsd:quoteChar="&quot;"/>
            <xsd:element name="LINE_INTERFACE_NUMBER"
type="xsd:string"
                        nxsd:style="terminated"
                        nxsd:terminatedBy=";"
```

```
            nxsd:quoteChar="&quot;"/>
        <xsd:element name="ITEM_CODE" type="xsd:string"
            nxsd:style="terminated"
            nxsd:terminatedBy=";"
            nxsd:quoteChar="&quot;"/>
        <xsd:element name="EAN_CODE" type="xsd:string"
            nxsd:style="terminated"
            nxsd:terminatedBy=";"
            nxsd:quoteChar="&quot;"/>
        <xsd:element name="SHORT_LABEL" type="xsd:string"
            nxsd:style="terminated"
            nxsd:terminatedBy=";"
            nxsd:quoteChar="&quot;"/>
        <xsd:element name="LONG_LABEL" type="xsd:string"
            nxsd:style="terminated"
            nxsd:terminatedBy=";"
            nxsd:quoteChar="&quot;"/>
        <xsd:element name="LINE_NUMBER" type="xsd:string"
            nxsd:style="terminated"
            nxsd:terminatedBy=";"
            nxsd:quoteChar="&quot;"/>
        <xsd:element name="MIN_ORDER_TYPE" type="xsd:string"
            nxsd:style="terminated"
            nxsd:terminatedBy=";"
            nxsd:quoteChar="&quot;"/>
        <xsd:element name="MIN_ORDER_UNIT" type="xsd:string"
```

```
                    nxsd:style="terminated"
                    nxsd:terminatedBy=";"
                    nxsd:quoteChar="&quot;"/>
          <xsd:element name="EXPECTED_QUANTITY"
type="xsd:string"
                    nxsd:style="terminated"
                    nxsd:terminatedBy=";"
                    nxsd:quoteChar="&quot;"/>
          <xsd:element name="ANNOUNCED_QUANTITY"
type="xsd:string"
                    nxsd:style="terminated"
                    nxsd:terminatedBy=";"
                    nxsd:quoteChar="&quot;"/>
          <xsd:element name="RECEIVED_QUANTITY"
type="xsd:string"
                    nxsd:style="terminated"
                    nxsd:terminatedBy=";"
                    nxsd:quoteChar="&quot;"/>
          <xsd:element name="STATUS" type="xsd:string"
                    nxsd:style="terminated"
                    nxsd:terminatedBy=";"
                    nxsd:quoteChar="&quot;"/>
          <xsd:element name="EXPIRY_DATE" type="xsd:string"
                    nxsd:style="terminated"
                    nxsd:terminatedBy=";"
                    nxsd:quoteChar="&quot;"/>
          <xsd:element name="LOT_NUMBER" type="xsd:string"
```

```
                    nxsd:style="terminated"
                    nxsd:terminatedBy=";"
                    nxsd:quoteChar="&quot;"/>
          <xsd:element name="OPERATION_CODE_1"
type="xsd:string"
                    nxsd:style="terminated"
                    nxsd:terminatedBy=";"
                    nxsd:quoteChar="&quot;"/>
          <xsd:element name="SUB_OPERATION_CODE_1"
type="xsd:string"
                    nxsd:style="terminated"
                    nxsd:terminatedBy=";"
                    nxsd:quoteChar="&quot;"/>
          <xsd:element name="BLOCKING_TYPE" type="xsd:string"
                    nxsd:style="terminated"
                    nxsd:terminatedBy=";"
                    nxsd:quoteChar="&quot;"/>
          <xsd:element name="BLOCKING_CODE" type="xsd:string"
                    nxsd:style="terminated"
                    nxsd:terminatedBy=";"
                    nxsd:quoteChar="&quot;"/>
          <xsd:element name="RESERVATION" type="xsd:string"
                    nxsd:style="terminated"
                    nxsd:terminatedBy=";"
                    nxsd:quoteChar="&quot;"/>
          <xsd:element name="RECIPIENT" type="xsd:string"
                    nxsd:style="terminated"
```

```
                    nxsd:terminatedBy=";"
                    nxsd:quoteChar="&quot;"/>
          <xsd:element name="UNIT_PRICE_EXCL_TAX"
type="xsd:string"
                    nxsd:style="terminated"
                    nxsd:terminatedBy=";"
                    nxsd:quoteChar="&quot;"/>
          <xsd:element name="CLIENT_SSCC" type="xsd:string"
                    nxsd:style="terminated"
                    nxsd:terminatedBy=";"
                    nxsd:quoteChar="&quot;"/>
          <xsd:element name="UNIT_QUANTITY" type="xsd:string"
                    nxsd:style="terminated"
                    nxsd:terminatedBy=";"
                    nxsd:quoteChar="&quot;"/>
          <xsd:element name="PACK_QUANTITY" type="xsd:string"
                    nxsd:style="terminated"
                    nxsd:terminatedBy=";"
                    nxsd:quoteChar="&quot;"/>
          <xsd:element name="CARTON_QUANTITY"
type="xsd:string"
                    nxsd:style="terminated"
                    nxsd:terminatedBy=";"
                    nxsd:quoteChar="&quot;"/>
          <xsd:element name="LAYER_QUANTITY" type="xsd:string"
                    nxsd:style="terminated"
                    nxsd:terminatedBy=";"
```

```
                nxsd:quoteChar="&quot;"/>
        <xsd:element name="PALLET_QUANTITY" type="xsd:string"
                nxsd:style="terminated"
                nxsd:terminatedBy=";"
                nxsd:quoteChar="&quot;"/>
        <xsd:element name="TRANSPORT_LINE" type="xsd:string"
                nxsd:style="terminated"
                nxsd:terminatedBy=";"
                nxsd:quoteChar="&quot;"/>
        <xsd:element name="REASON" type="xsd:string"
                nxsd:style="terminated"
                nxsd:terminatedBy="${eol}"
                nxsd:quoteChar="&quot;"/>
      </xsd:sequence>
     </xsd:complexType>
    </xsd:element>
   </xsd:choice>
  </xsd:complexType>
 </xsd:element>
</xsd:schema>
```

##

1.3. Resulting XML Structure

After parsing, OIC produces the following XML structure :

<RCV>

<Header>
<WAREHOUSE_CODE>WAREHOUSE_A</WAREHOUSE_CODE>

<RECEIPT_ID>REC001</RECEIPT_ID>
<PO_NUMBER>PO_001</PO_NUMBER>

...

</Header>

<Line>

<RECEIPT_ID>REC001</RECEIPT_ID>
<LINE_NUMBER>1</LINE_NUMBER>

...

</Line>

<Header>
<WAREHOUSE_CODE>WAREHOUSE_B</WAREHOUSE_CODE>
<RECEIPT_ID>REC002</RECEIPT_ID>
<PO_NUMBER>PO_002</PO_NUMBER>

...

</Header>

<Line>

<RECEIPT_ID>REC002</RECEIPT_ID>
<LINE_NUMBER>1</LINE_NUMBER>

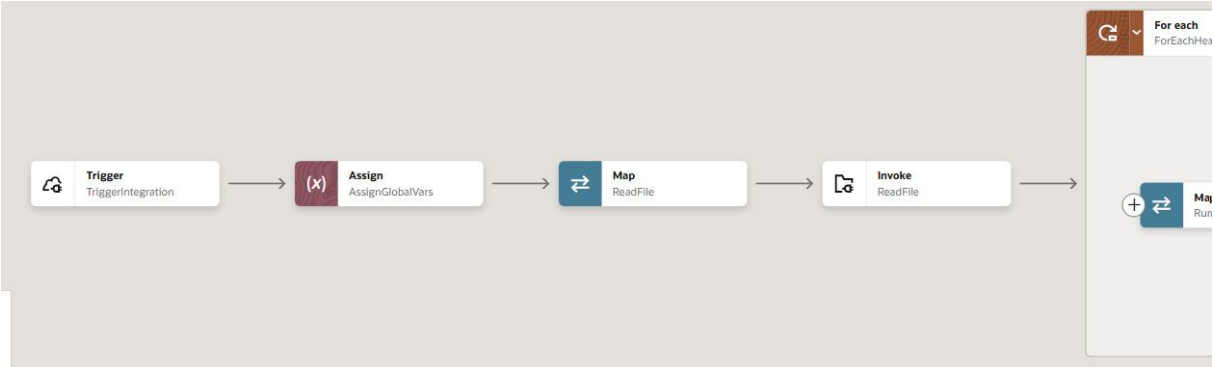
...

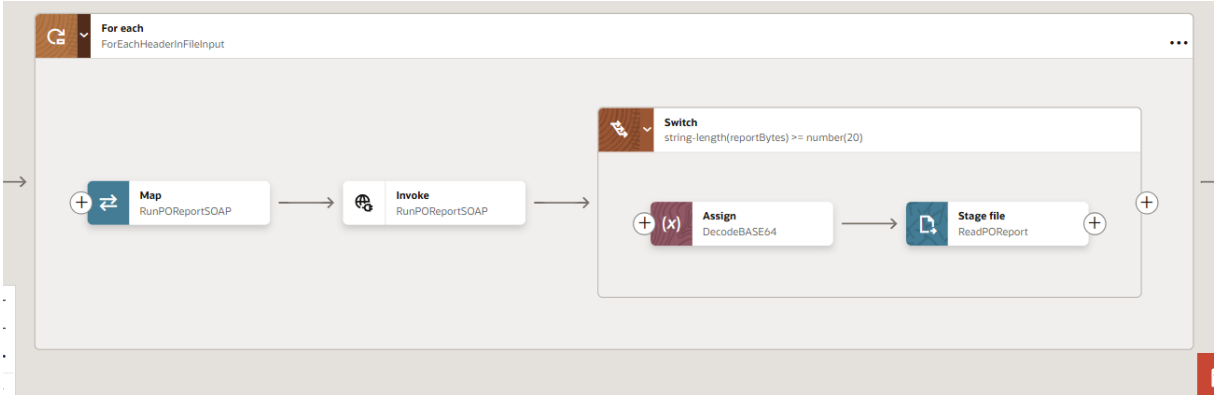
</Line>

</RCV>

2. POC1 : For Each Approach

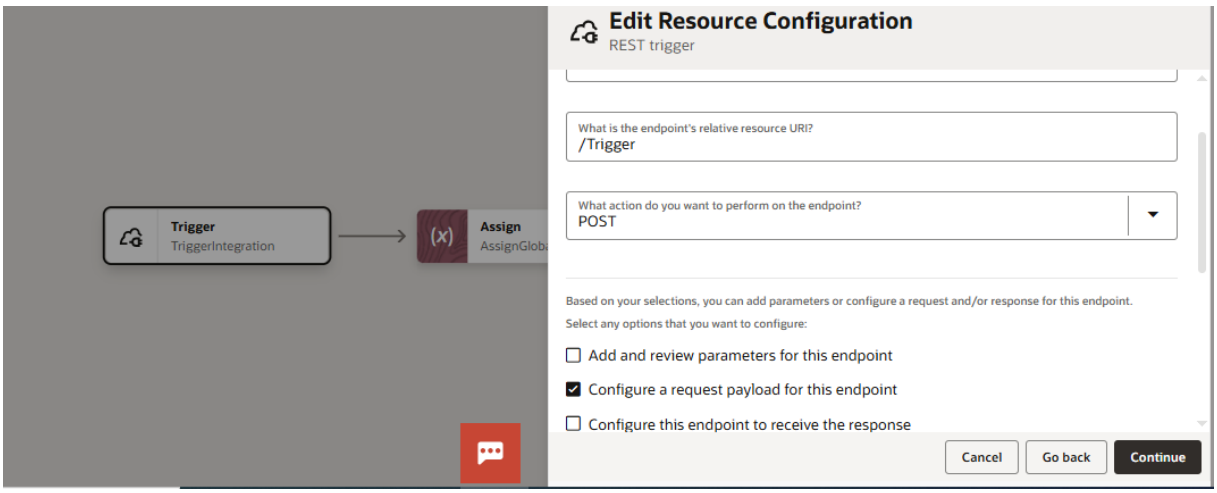
2.1. Integration Flow

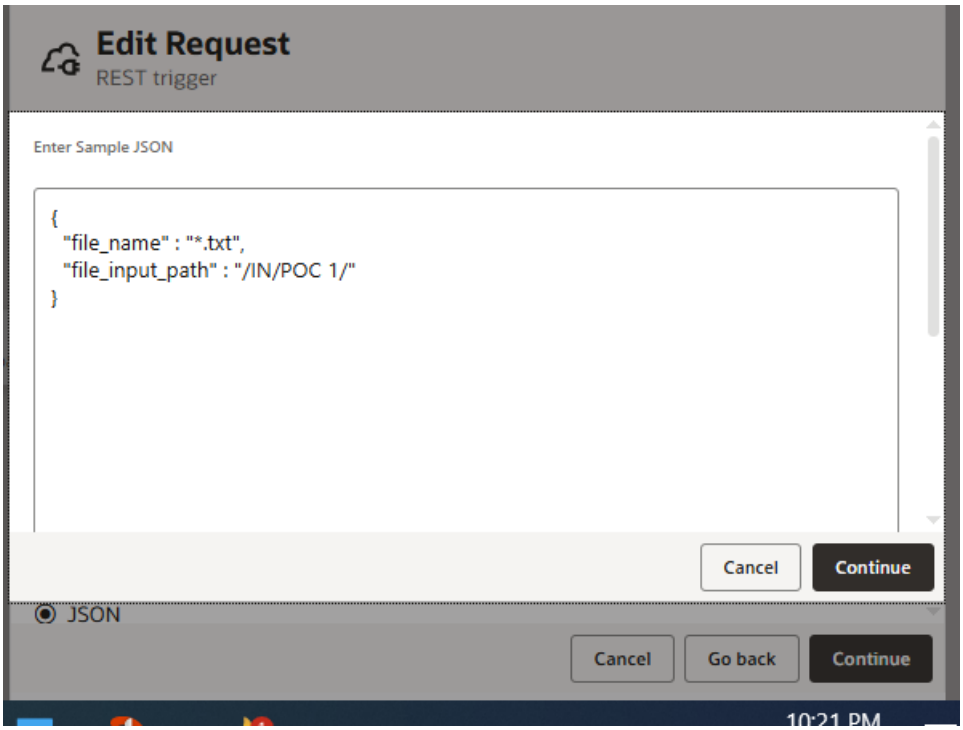




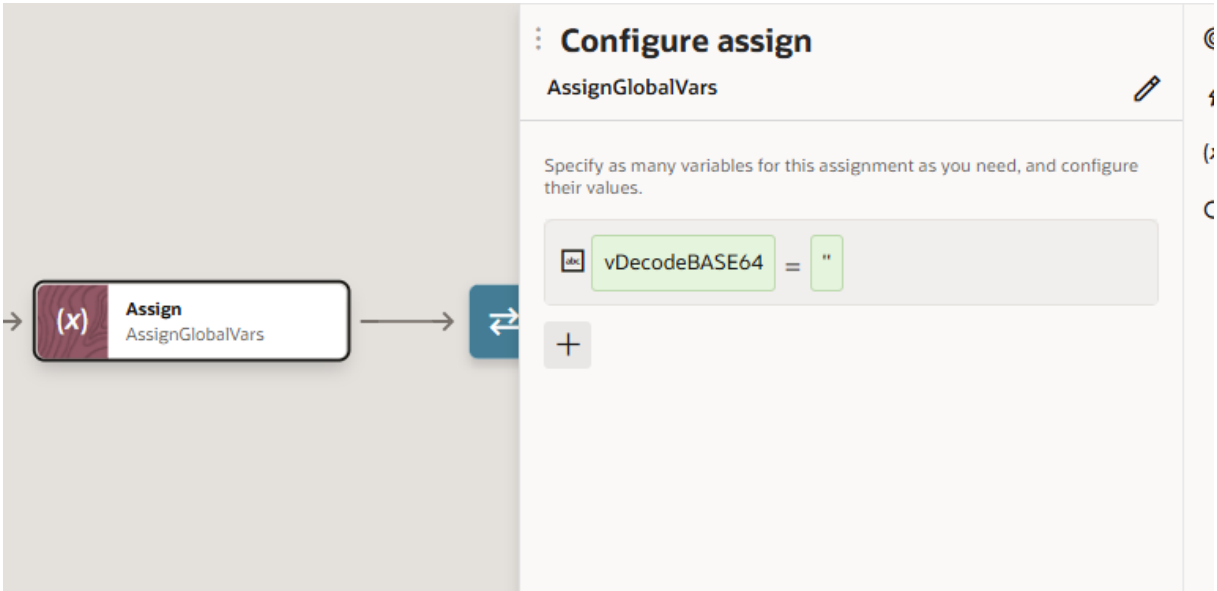
The integration flow is structured as follows :

Trigger :

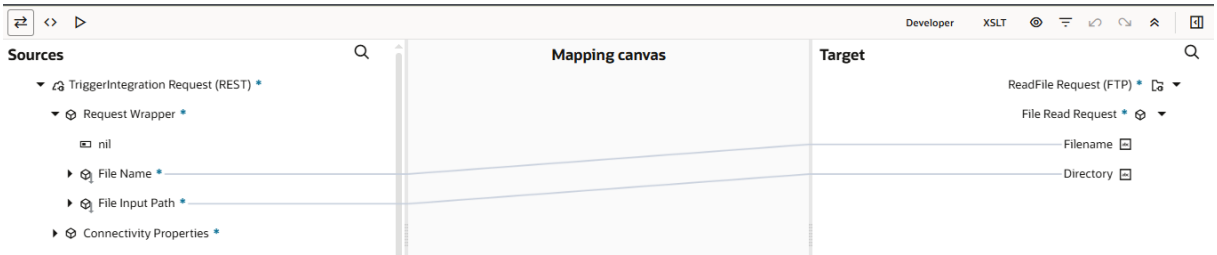




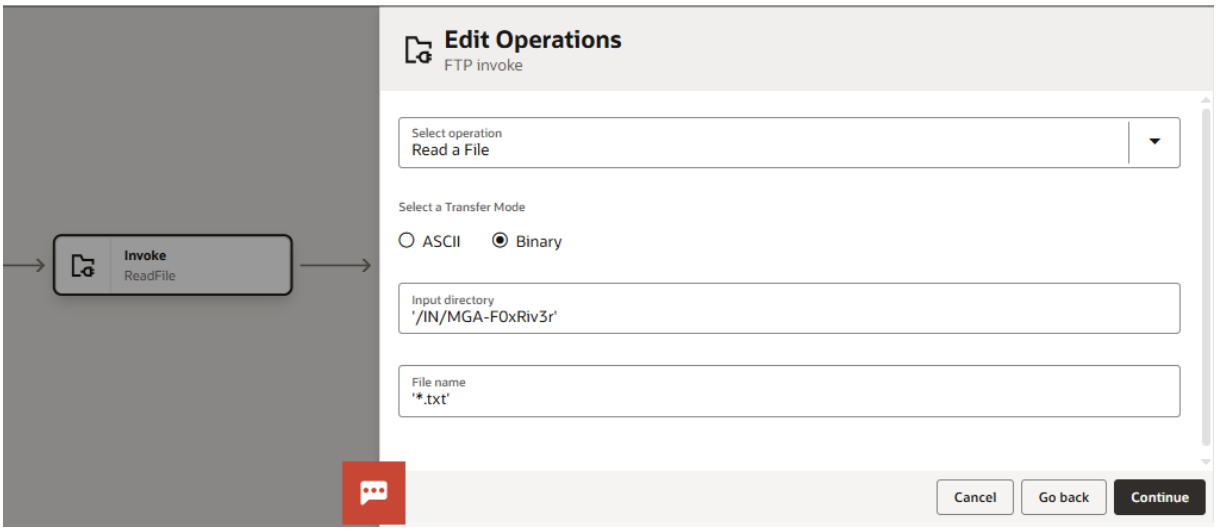
AssignGlobalVars :

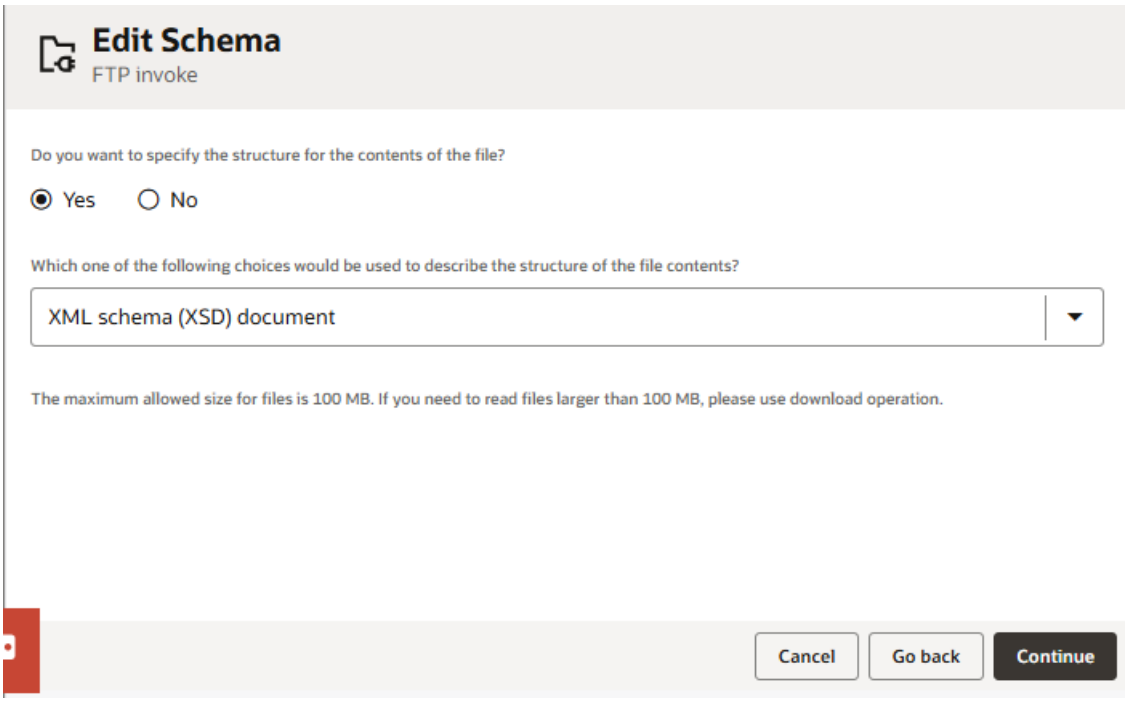


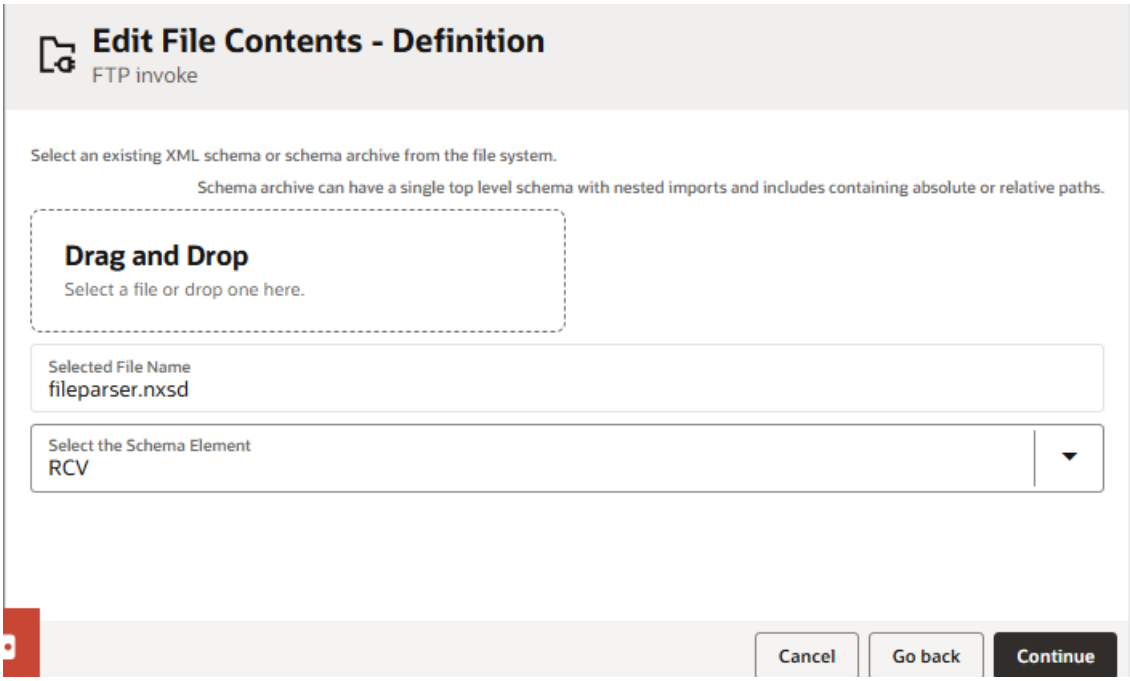
Map ReadFile : mapper to configure the file read



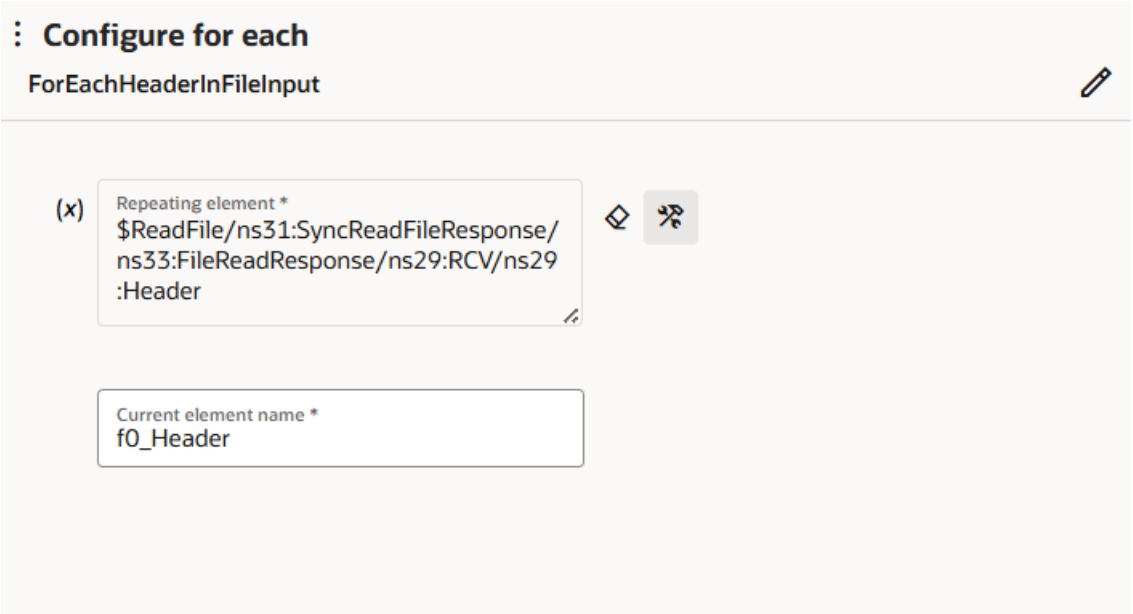
Invoke ReadFile : the flat file is read and parsed using the NXSD schema



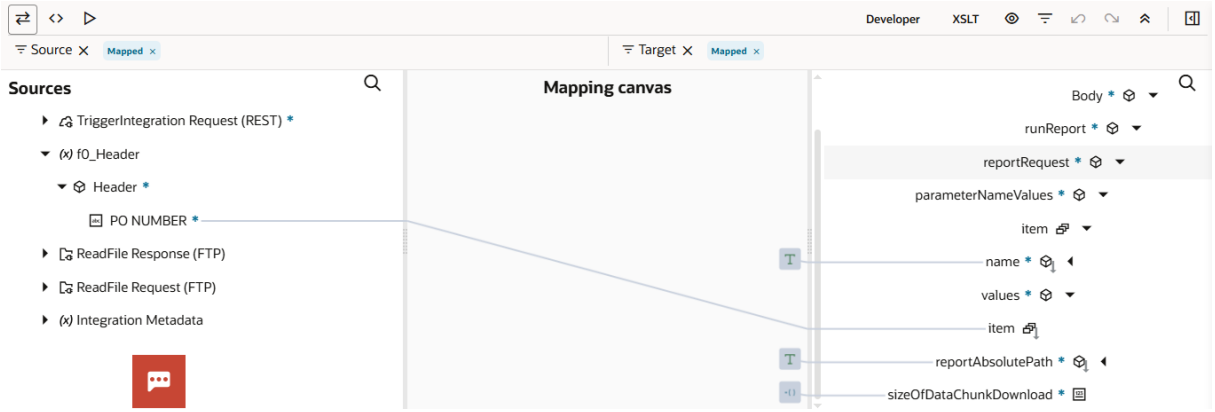


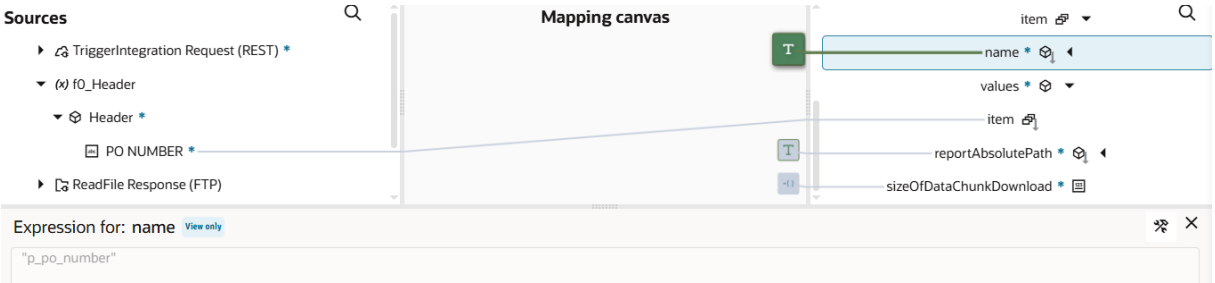


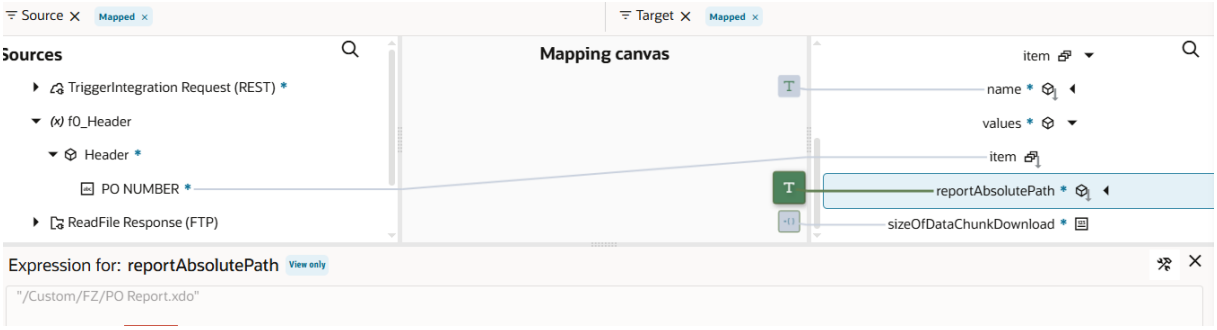
For Each (ForEachHeaderInFileInput) : loops over each Header in the parsed XML

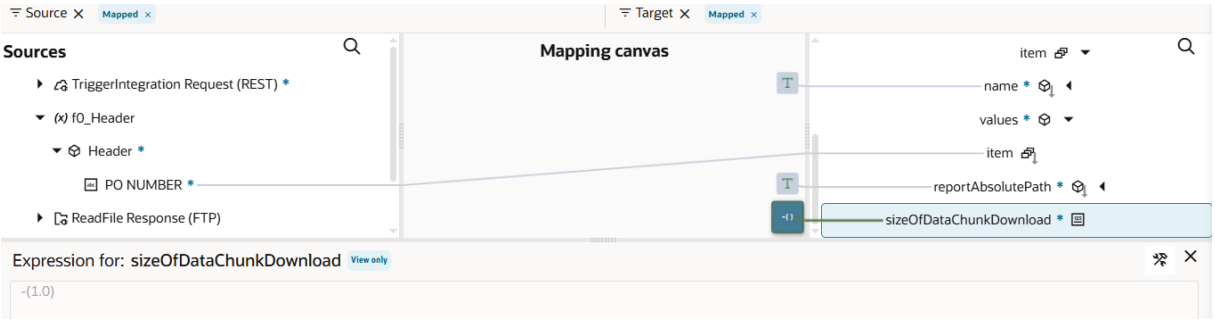


Map RunPOReportSOAP : maps a single PO_NUMBER to the BIP report parameter

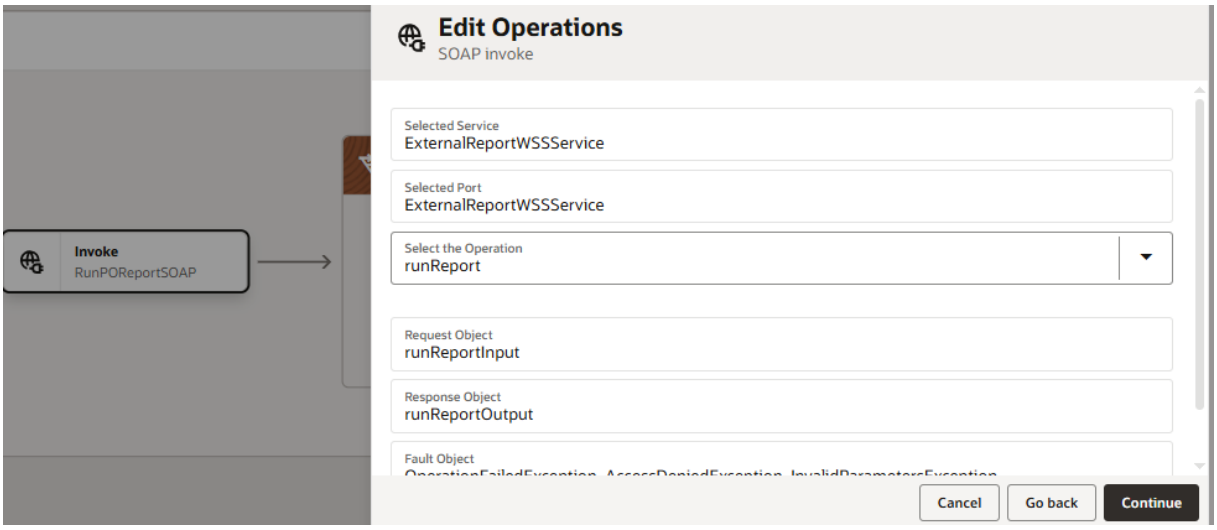




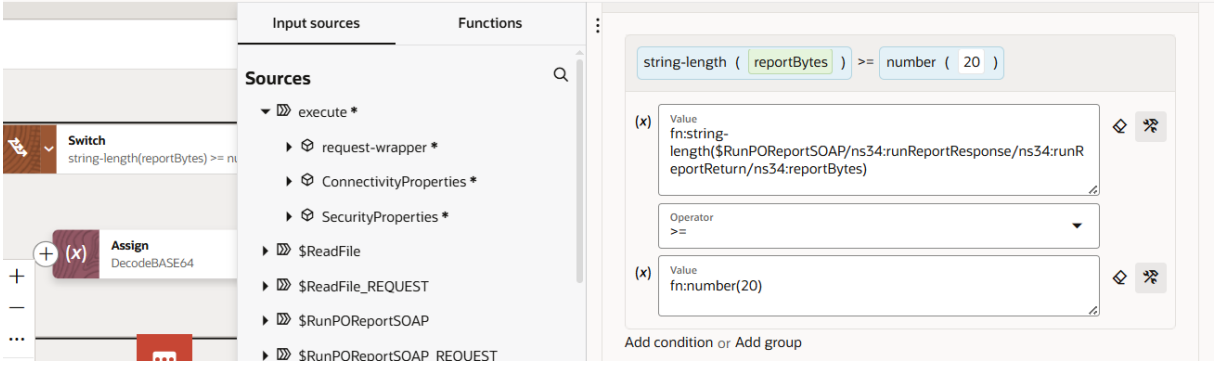




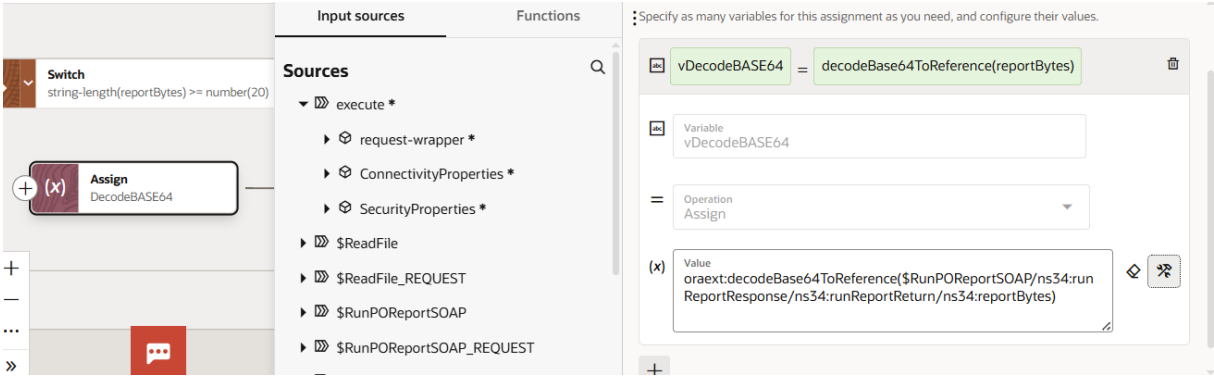
Invoke RunPOReportSOAP : calls the BIP report with one PO Number



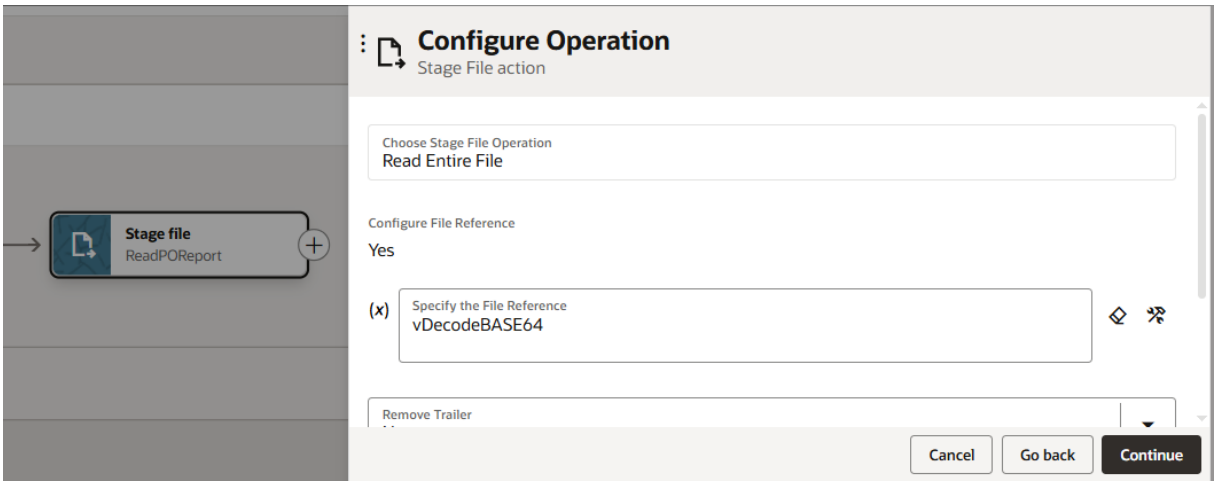
Switch : checks if reportBytes length >= 20 (report returned data)



Assign DecodeBase64 : decodes the report bytes



Stage File ReadPOReport : reads the decoded report



BIP Parameter Mapping (POC1 XSLT) In this approach, a single PO_NUMBER is passed per iteration :

```
<ns36:parameterNameValues xml:id="id_69">
      <ns36:item xml:id="id_70">
            <ns36:name xml:id="id_71">
                  <xsl:value-of xml:id="id_237" select="'p_po_number'"/>
            </ns36:name>
            <ns36:values xml:id="id_73">
                  <ns36:item xml:id="id_275">
                        <xsl:value-of xml:id="id_301" select="$f0_Header/nsmpr4:Header/nsmpr4:PO_NUMBER"/>
                  </ns36:item>
            </ns36:values>
      </ns36:item>
</ns36:parameterNameValues>
```

BIP SQL Query (POC1) The SQL query uses a simple equality filter:
WHERE PO_DATA.SEGMENT1 = :p_po_number

```
##########################################

WITH PO_DATA AS (
        SELECT PurchasingDocHDR.CURRENCY_CODE,
          PODistribution.PO_DISTRIBUTION_ID,
          PODistribution.CODE_COMBINATION_ID,
          PODistribution.DELIVER_TO_LOCATION_ID,
          PODistribution.DESTINATION_ORGANIZATION_ID,
          PurchasingDocShipment.LINE_LOCATION_ID,
          PurchasingDocShipment.SHIP_TO_LOCATION_ID,
          PurchasingDocLine.ITEM_DESCRIPTION,
          PurchasingDocLine.ITEM_ID,
          PurchasingDocLine.PO_LINE_ID,
          PurchasingDocLine.REQ_BU_ID,
          PurchasingDocLine.VENDOR_PRODUCT_NUM,
          PurchasingDocHDR.BILL_TO_LOCATION_ID,
          PurchasingDocHDR.BILLTO_BU_ID,
          PurchasingDocHDR.PO_HEADER_ID,
          PurchasingDocHDR.PRC_BU_ID,
```

```
          PurchasingDocHDR.SEGMENT1,
          PurchasingDocHDR.SOLDTO_LE_ID,
          PurchasingDocHDR.VENDOR_ID,
          PurchasingDocHDR.VENDOR_SITE_ID,
          PurchasingSysParam.INVENTORY_ORGANIZATION_ID,
          Category.CATEGORY_ID AS C_CATEGORY_ID,
          Category.CATEGORY_NAME AS C_CATEGORY_NAME,
          PurchasingDocShipment.SHIP_TO_CUST_LOCATION_ID,
          PODistribution.DELIVER_TO_CUST_LOCATION_ID,
          PurchasingDocLine.LINE_NUM LINE_NUMBER,
          PODistribution.DESTINATION_SUBINVENTORY
        FROM
          PO_DISTRIBUTIONS_ALL PODistribution,
          PO_LINE_LOCATIONS_ALL PurchasingDocShipment,
          PO_LINES_ALL PurchasingDocLine,
          PO_HEADERS_ALL PurchasingDocHDR,
          PO_SYSTEM_PARAMETERS_ALL PurchasingSysParam,
          EGP_CATEGORIES_VL Category
        WHERE

            PODistribution.LINE_LOCATION_ID =
PurchasingDocShipment.LINE_LOCATION_ID
            AND PurchasingDocShipment.PO_LINE_ID =
PurchasingDocLine.PO_LINE_ID
            AND PurchasingDocLine.PO_HEADER_ID =
PurchasingDocHDR.PO_HEADER_ID
            AND PurchasingDocHDR.PRC_BU_ID =
PurchasingSysParam.PRC_BU_ID
            AND PurchasingDocLine.CATEGORY_ID =
Category.CATEGORY_ID(+)
            AND PurchasingDocHDR.TYPE_LOOKUP_CODE = 'STANDARD'

      ),

q1 AS (
SELECT
          SupplierSite.VENDOR_SITE_ID,  SupplierSite.VENDOR_SITE_COD
E,
          Supplier.VENDOR_ID AS VENDOR_ID2
        FROM
          POZ_SUPPLIER_SITES_ALL_M SupplierSite,
```

```
          POZ_SUPPLIERS Supplier
        WHERE
          SupplierSite.VENDOR_ID = Supplier.VENDOR_ID(+)
),
q2 AS (
SELECT
          Supplier.SEGMENT1,
          Supplier.VENDOR_ID AS VENDOR_ID1
        FROM
          POZ_SUPPLIERS Supplier)
,
q4 AS (
 SELECT
          OrgUnit.ORGANIZATION_ID AS ORGANIZATION_ID1,
          OrgUnit.EFFECTIVE_START_DATE AS EFFECTIVE_START_DATE3,
          OrgUnit.EFFECTIVE_END_DATE AS EFFECTIVE_END_DATE3,
          OrgInfo.ORG_INFORMATION_ID,
          OrgUnitTl.NAME,
          OrgUnitTl.ORGANIZATION_ID,
          OrgUnitTl.EFFECTIVE_START_DATE,
          OrgUnitTl.EFFECTIVE_END_DATE,
          OrgUnitTl.LANGUAGE
        FROM
          HR_ALL_ORGANIZATION_UNITS_F OrgUnit,
          HR_ORGANIZATION_INFORMATION_F OrgInfo,
          HR_ORGANIZATION_UNITS_F_TL OrgUnitTl
        WHERE
          OrgUnit.ORGANIZATION_ID = OrgInfo.ORGANIZATION_ID
          AND ('FUN_BUSINESS_UNIT') =
OrgInfo.ORG_INFORMATION_CONTEXT
          AND OrgUnit.ORGANIZATION_ID = OrgUnitTl.ORGANIZATION_ID
          AND OrgUnit.EFFECTIVE_START_DATE =
OrgUnitTl.EFFECTIVE_START_DATE
          AND OrgUnit.EFFECTIVE_END_DATE =
OrgUnitTl.EFFECTIVE_END_DATE
          AND (
            USERENV('LANG')
          ) = OrgUnitTl.LANGUAGE
          AND (
            SYSDATE BETWEEN OrgUnit.EFFECTIVE_START_DATE
            AND OrgUnit.EFFECTIVE_END_DATE
```

```
          )
          AND (
            SYSDATE BETWEEN OrgInfo.EFFECTIVE_START_DATE
            AND OrgInfo.EFFECTIVE_END_DATE
          )
          AND (
            SYSDATE BETWEEN OrgUnitTl.EFFECTIVE_START_DATE
            AND OrgUnitTl.EFFECTIVE_END_DATE
          )




),
q5 AS (
SELECT
          Supplier.SEGMENT1,
          Supplier.VENDOR_ID AS VENDOR_ID1,
          Party.PARTY_ID AS PARTY_ID1,
          Party.PARTY_NAME AS PARTY_NAME1
        FROM
          POZ_SUPPLIERS Supplier,
          HZ_PARTIES Party
        WHERE
          (
            Supplier.PARTY_ID = Party.PARTY_ID
          )


), q6 AS (
SELECT
          LegalEntity.LEGAL_ENTITY_ID,
          LegalEntity.NAME,
LegalEntity.LEGAL_ENTITY_IDENTIFIER AS SoldToLE
        FROM
          XLE_ENTITY_PROFILES LegalEntity
),
q7 AS (
SELECT
          OrgUnit.ORGANIZATION_ID AS ORGANIZATION_ID1,
          OrgUnit.EFFECTIVE_START_DATE AS EFFECTIVE_START_DATE3,
```

```
          OrgUnit.EFFECTIVE_END_DATE AS EFFECTIVE_END_DATE3,
          OrgInfo.ORG_INFORMATION_ID,
          OrgUnitTl.NAME,
          OrgUnitTl.ORGANIZATION_ID,
          OrgUnitTl.EFFECTIVE_START_DATE,
          OrgUnitTl.EFFECTIVE_END_DATE,
          OrgUnitTl.LANGUAGE
        FROM
          HR_ALL_ORGANIZATION_UNITS_F OrgUnit,
          HR_ORGANIZATION_INFORMATION_F OrgInfo,
          HR_ORGANIZATION_UNITS_F_TL OrgUnitTl
        WHERE
          OrgUnit.ORGANIZATION_ID = OrgInfo.ORGANIZATION_ID
          AND ('FUN_BUSINESS_UNIT') =
OrgInfo.ORG_INFORMATION_CONTEXT
          AND OrgUnit.ORGANIZATION_ID = OrgUnitTl.ORGANIZATION_ID
          AND OrgUnit.EFFECTIVE_START_DATE =
OrgUnitTl.EFFECTIVE_START_DATE
          AND OrgUnit.EFFECTIVE_END_DATE =
OrgUnitTl.EFFECTIVE_END_DATE
          AND (
            USERENV('LANG')
          ) = OrgUnitTl.LANGUAGE
          AND (
            SYSDATE BETWEEN OrgUnit.EFFECTIVE_START_DATE
            AND OrgUnit.EFFECTIVE_END_DATE
          )
          AND (
            SYSDATE BETWEEN OrgInfo.EFFECTIVE_START_DATE
            AND OrgInfo.EFFECTIVE_END_DATE
          )
          AND (
            SYSDATE BETWEEN OrgUnitTl.EFFECTIVE_START_DATE
            AND OrgUnitTl.EFFECTIVE_END_DATE
          )
),
q8 AS (
SELECT
          LocationDetailsTl.LOCATION_DETAILS_ID AS
LOCATION_DETAILS_ID1,
          LocationDetailsTl.EFFECTIVE_START_DATE AS
```

```
EFFECTIVE_START_DATE1,
          LocationDetailsTl.EFFECTIVE_END_DATE AS
EFFECTIVE_END_DATE1,
          LocationDetailsTl.LANGUAGE,
          LocationDetailsTl.LOCATION_CODE,
          LocationDetailsTl.LOCATION_NAME,
          LocationDetails.LOCATION_DETAILS_ID AS
LOCATION_DETAILS_ID1,
          LocationDetails.EFFECTIVE_START_DATE AS
EFFECTIVE_START_DATE2,
          LocationDetails.EFFECTIVE_END_DATE AS
EFFECTIVE_END_DATE2,
          LocationDetails.LOCATION_ID,
          LocationPEO.LOCATION_ID AS LOCATION_ID373,
          LocationPEO.INTERNAL_LOCATION_CODE
        FROM
          PER_LOCATION_DETAILS_F_VL LocationDetails,
          PER_LOCATION_DETAILS_F_TL LocationDetailsTl,
          PER_LOCATIONS LocationPEO
        WHERE
          LocationDetails.LOCATION_DETAILS_ID =
LocationDetailsTl.LOCATION_DETAILS_ID
          AND LocationDetails.EFFECTIVE_START_DATE =
LocationDetailsTl.EFFECTIVE_START_DATE
          AND LocationDetails.EFFECTIVE_END_DATE =
LocationDetailsTl.EFFECTIVE_END_DATE
          AND LocationDetails.LOCATION_ID = LocationPEO.LOCATION_ID
          AND (
            SYSDATE BETWEEN LocationDetails.EFFECTIVE_START_DATE
            AND LocationDetails.EFFECTIVE_END_DATE
          )
          AND (
            SYSDATE BETWEEN LocationDetailsTl.EFFECTIVE_START_DATE
            AND LocationDetailsTl.EFFECTIVE_END_DATE
          )




),
q9 AS (
```

```
SELECT
          OrgUnit.ORGANIZATION_ID AS ORGANIZATION_ID1,
          OrgUnit.EFFECTIVE_START_DATE AS EFFECTIVE_START_DATE3,
          OrgUnit.EFFECTIVE_END_DATE AS EFFECTIVE_END_DATE3,
          OrgInfo.ORG_INFORMATION_ID,
          OrgUnitTl.NAME,
          OrgUnitTl.ORGANIZATION_ID,
          OrgUnitTl.EFFECTIVE_START_DATE,
          OrgUnitTl.EFFECTIVE_END_DATE,
          OrgUnitTl.LANGUAGE
        FROM
          HR_ALL_ORGANIZATION_UNITS_F OrgUnit,
          HR_ORGANIZATION_INFORMATION_F OrgInfo,
          HR_ORGANIZATION_UNITS_F_TL OrgUnitTl
        WHERE
          OrgUnit.ORGANIZATION_ID = OrgInfo.ORGANIZATION_ID
          AND ('FUN_BUSINESS_UNIT') =
OrgInfo.ORG_INFORMATION_CONTEXT
          AND OrgUnit.ORGANIZATION_ID = OrgUnitTl.ORGANIZATION_ID
          AND OrgUnit.EFFECTIVE_START_DATE =
OrgUnitTl.EFFECTIVE_START_DATE
          AND OrgUnit.EFFECTIVE_END_DATE =
OrgUnitTl.EFFECTIVE_END_DATE
          AND (
            USERENV('LANG')
          ) = OrgUnitTl.LANGUAGE
          AND (
            SYSDATE BETWEEN OrgUnit.EFFECTIVE_START_DATE
            AND OrgUnit.EFFECTIVE_END_DATE
          )
          AND (
            SYSDATE BETWEEN OrgInfo.EFFECTIVE_START_DATE
            AND OrgInfo.EFFECTIVE_END_DATE
          )
          AND (
            SYSDATE BETWEEN OrgUnitTl.EFFECTIVE_START_DATE
            AND OrgUnitTl.EFFECTIVE_END_DATE
          )


), q10 AS (
```

```
         SELECT  CodeCombination.CODE_COMBINATION_ID
        FROM
          GL_CODE_COMBINATIONS CodeCombination


), q11 AS (
SELECT

          PO_HDR.PO_HEADER_ID AS po_hdr_id,
          PO_HDR.SH_ATTRIBUTE1,
          PO_HDR.SH_ATTRIBUTE2,
          PO_HDR.SH_ATTRIBUTE_DATE1
        FROM
          PO_HEADERS_ALL PO_HDR)
, q12 AS (
        SELECT
          LocationDetailsTl.LOCATION_DETAILS_ID AS
LOCATION_DETAILS_ID1,
          LocationDetailsTl.EFFECTIVE_START_DATE AS
EFFECTIVE_START_DATE1,
          LocationDetailsTl.EFFECTIVE_END_DATE AS
EFFECTIVE_END_DATE1,
          LocationDetailsTl.LANGUAGE,
          LocationDetailsTl.LOCATION_CODE,
          LocationDetails.LOCATION_DETAILS_ID AS
LOCATION_DETAILS_ID1,
          LocationDetails.EFFECTIVE_START_DATE AS
EFFECTIVE_START_DATE2,
          LocationDetails.EFFECTIVE_END_DATE AS
EFFECTIVE_END_DATE2,
          LocationDetails.LOCATION_ID
        FROM
          PER_LOCATION_DETAILS_F_VL LocationDetails,
          PER_LOCATION_DETAILS_F_TL LocationDetailsTl
        WHERE
          LocationDetails.LOCATION_DETAILS_ID =
LocationDetailsTl.LOCATION_DETAILS_ID
          AND LocationDetails.EFFECTIVE_START_DATE =
LocationDetailsTl.EFFECTIVE_START_DATE
          AND LocationDetails.EFFECTIVE_END_DATE =
LocationDetailsTl.EFFECTIVE_END_DATE
```

```
          AND (
            SYSDATE BETWEEN LocationDetails.EFFECTIVE_START_DATE
            AND LocationDetails.EFFECTIVE_END_DATE
          )
          AND (
            SYSDATE BETWEEN LocationDetailsTl.EFFECTIVE_START_DATE
            AND LocationDetailsTl.EFFECTIVE_END_DATE
          )
), q13 AS (
SELECT
          LocationDetailsTl.LOCATION_DETAILS_ID AS
LOCATION_DETAILS_ID1,
          LocationDetailsTl.EFFECTIVE_START_DATE AS
EFFECTIVE_START_DATE1,
          LocationDetailsTl.EFFECTIVE_END_DATE AS
EFFECTIVE_END_DATE1,
          LocationDetailsTl.LANGUAGE,
          LocationDetailsTl.LOCATION_CODE,
          LocationDetails.LOCATION_DETAILS_ID AS
LOCATION_DETAILS_ID1,
          LocationDetails.EFFECTIVE_START_DATE AS
EFFECTIVE_START_DATE2,
          LocationDetails.EFFECTIVE_END_DATE AS
EFFECTIVE_END_DATE2,
          LocationDetails.LOCATION_ID
        FROM
          PER_LOCATION_DETAILS_F_VL LocationDetails,
          PER_LOCATION_DETAILS_F_TL LocationDetailsTl
        WHERE
          LocationDetails.LOCATION_DETAILS_ID =
LocationDetailsTl.LOCATION_DETAILS_ID
          AND LocationDetails.EFFECTIVE_START_DATE =
LocationDetailsTl.EFFECTIVE_START_DATE
          AND LocationDetails.EFFECTIVE_END_DATE =
LocationDetailsTl.EFFECTIVE_END_DATE
          AND (
            SYSDATE BETWEEN LocationDetails.EFFECTIVE_START_DATE
            AND LocationDetails.EFFECTIVE_END_DATE
          )
          AND (
            SYSDATE BETWEEN LocationDetailsTl.EFFECTIVE_START_DATE
```

```
            AND LocationDetailsTl.EFFECTIVE_END_DATE
          )
), q14 AS
(
        SELECT
          esib.INVENTORY_ITEM_ID,
          esib.ORGANIZATION_ID AS ORGANIZATION_ID1,
          esib.SOURCE_ORGANIZATION_ID AS
SOURCE_ORGANIZATION_ID1,
          OrgUnitTLF.EFFECTIVE_END_DATE AS EFFECTIVE_END_DATE1,
          OrgUnitTLF.EFFECTIVE_START_DATE AS
EFFECTIVE_START_DATE1,
          OrgUnitTLF.LANGUAGE AS LANGUAGE1,
          OrgUnitTLF.NAME AS NAME2800,
          OrgUnitTLF.ORGANIZATION_ID AS ORGANIZATION_ID3
        FROM
          EGP_SYSTEM_ITEMS_B_V esib,
          HR_ALL_ORGANIZATION_UNITS_F OrgUnitF,
          HR_ORGANIZATION_UNITS_F_TL OrgUnitTLF
        WHERE
          (
            esib.ORGANIZATION_ID = OrgUnitF.ORGANIZATION_ID
            AND OrgUnitF.ORGANIZATION_ID =
OrgUnitTLF.ORGANIZATION_ID
            AND OrgUnitF.EFFECTIVE_START_DATE =
OrgUnitTLF.EFFECTIVE_START_DATE
            AND OrgUnitF.EFFECTIVE_END_DATE =
OrgUnitTLF.EFFECTIVE_END_DATE
            AND (
              USERENV('LANG')
            ) = OrgUnitTLF.LANGUAGE
            AND (
              SYSDATE BETWEEN OrgUnitF.EFFECTIVE_START_DATE
              AND OrgUnitF.EFFECTIVE_END_DATE
            )
            AND (
              SYSDATE BETWEEN OrgUnitTLF.EFFECTIVE_START_DATE
              AND OrgUnitTLF.EFFECTIVE_END_DATE
            )
          )
          --AND (((esib.TEMPLATE_ITEM_FLAG = 'N')))
```

```
      ) , q15 AS
      (
        SELECT
          esib.INVENTORY_ITEM_ID,
          esib.ITEM_NUMBER,
          esib.ORGANIZATION_ID AS ORGANIZATION_ID1
        FROM
          EGP_SYSTEM_ITEMS_B_V esib
        WHERE 1=1
          --AND ((esib.TEMPLATE_ITEM_FLAG = 'N'))
      ), q16 AS
(
        SELECT
          OrgParam.ORGANIZATION_CODE,
          OrgParam.ORGANIZATION_ID
        FROM
          INV_ORG_PARAMETERS OrgParam
      ) ,
q17 AS (

SELECT
PO_DATA.DESTINATION_SUBINVENTORY,
q6.SoldToLE,
PO_DATA.CURRENCY_CODE,
q1.VENDOR_SITE_CODE,
q12.LOCATION_CODE DELIVER_TO_LOCATION,
q13.LOCATION_CODE SHIP_TO_LOCATION,
q4.NAME BILLTO_BU,
q8.LOCATION_CODE BILL_TO_LOCATION_CODE,
q8.LOCATION_NAME BILL_TO_LOCATION_NAME,
q9.NAME PRC_BU,
PO_DATA.SEGMENT1 PO_NUMBER,
q7.NAME REQ_BU,
q6.NAME LEGAL_ENTITY,
q5.PARTY_NAME1 SUPPLIER_NAME,
q5.SEGMENT1 SUPPLIER_NUMBER,
PO_DATA.ITEM_DESCRIPTION ITEM_DESCRIPTION,
q15.ITEM_NUMBER, PO_DATA.LINE_NUMBER,
q16.ORGANIZATION_CODE DESTINATION_ORGANIZATION,
ROW_NUMBER() OVER (
          PARTITION BY PO_DATA.PO_LINE_ID
```

```
          ORDER BY PO_DATA.LINE_NUMBER
       ) rn

FROM PO_DATA,
q4, q5,
q6, q1, q2, q7,
q8, q9, q10, q11,
q12, q13, q15, q14, q16
WHERE 1=1

        AND PO_DATA.BILLTO_BU_ID = q4.ORGANIZATION_ID1(+)
        AND PO_DATA.VENDOR_ID = q5.VENDOR_ID1(+)
        AND PO_DATA.ITEM_ID = q14.INVENTORY_ITEM_ID(+)
        AND PO_DATA.INVENTORY_ORGANIZATION_ID =
q14.ORGANIZATION_ID1(+)
        AND PO_DATA.SOLDTO_LE_ID = q6.LEGAL_ENTITY_ID(+)
        AND PO_DATA.VENDOR_SITE_ID = q1.VENDOR_SITE_ID(+)
        AND q1.VENDOR_ID2 = q2.VENDOR_ID1(+)
        AND PO_DATA.DESTINATION_ORGANIZATION_ID =
q16.ORGANIZATION_ID(+)
        AND PO_DATA.REQ_BU_ID = q7.ORGANIZATION_ID1(+)
        AND PO_DATA.BILL_TO_LOCATION_ID = q8.LOCATION_ID(+)
        AND PO_DATA.PRC_BU_ID = q9.ORGANIZATION_ID1(+)
        AND PO_DATA.CODE_COMBINATION_ID =
q10.CODE_COMBINATION_ID(+)
        AND PO_DATA.PO_HEADER_ID = q11.po_hdr_id
        AND PO_DATA.ITEM_ID = q15.INVENTORY_ITEM_ID(+)
        AND PO_DATA.INVENTORY_ORGANIZATION_ID =
q15.ORGANIZATION_ID1(+)
        AND PO_DATA.DELIVER_TO_LOCATION_ID =
q12.LOCATION_ID(+)
        AND PO_DATA.SHIP_TO_LOCATION_ID = q13.LOCATION_ID(+)
 AND PO_DATA.SEGMENT1 = :p_PO_NUMBER

)
SELECT
DESTINATION_SUBINVENTORY,
SoldToLE,
CURRENCY_CODE,
VENDOR_SITE_CODE,
DELIVER_TO_LOCATION,
```

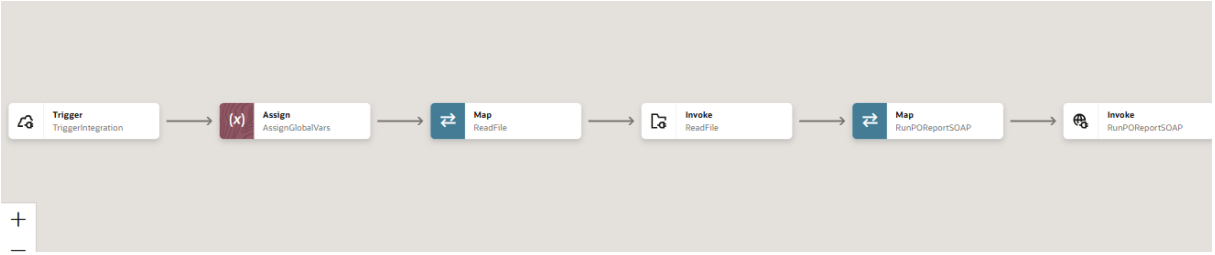
SHIP_TO_LOCATION,
BILLTO_BU,
BILL_TO_LOCATION_CODE,
BILL_TO_LOCATION_NAME,
PRC_BU,
PO_NUMBER,
REQ_BU,
LEGAL_ENTITY,
SUPPLIER_NAME,
SUPPLIER_NUMBER,
ITEM_DESCRIPTION,
ITEM_NUMBER,LINE_NUMBER,
DESTINATION_ORGANIZATION
 FROM q17 WHERE rn = '1' ORDER BY PO_NUMBER DESC

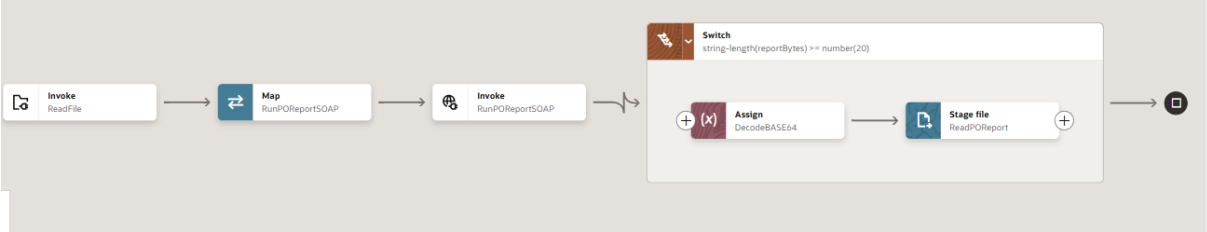
##

Limitation : For a file containing N headers, the integration makes N BIP calls. This is functional but not optimal from a performance standpoint.

3. POC2 : XSLT Optimization

The For Each loop is completely eliminated. The integration flow becomes :





The Challenge: The BIP parameter p_po_number expects a single value. To pass multiple values we need to :

1. **Build a comma-separated string from all PO_NUMBER values in the parsed XML**

2. **Pass it as a single string parameter : "PO_001,PO_002,PO_003"**
3. **Make the SQL query split this string back into individual values**

XSLT Solution — Building the CSV String Instead of passing one PO Number per iteration, the XSLT for-each iterates over all Headers and builds a single string :

```
##########################################

<ns36:item xml:id="id_70">

    <ns36:name xml:id="id_71">

    <xsl:value-of xml:id="id_237" select="'p_po_number'"/>

    </ns36:name>

   <ns36:values xml:id="id_73">

            <ns36:item>

        <xsl:for-each
select="$ReadFile/nsmpr3:SyncReadFileResponse/ns30:FileRead
Response/nsmpr4:RCV/nsmpr4:Header">

   <xsl:if test="position() &gt; 1">,</xsl:if>

    <xsl:value-of select="nsmpr4:PO_NUMBER"/>

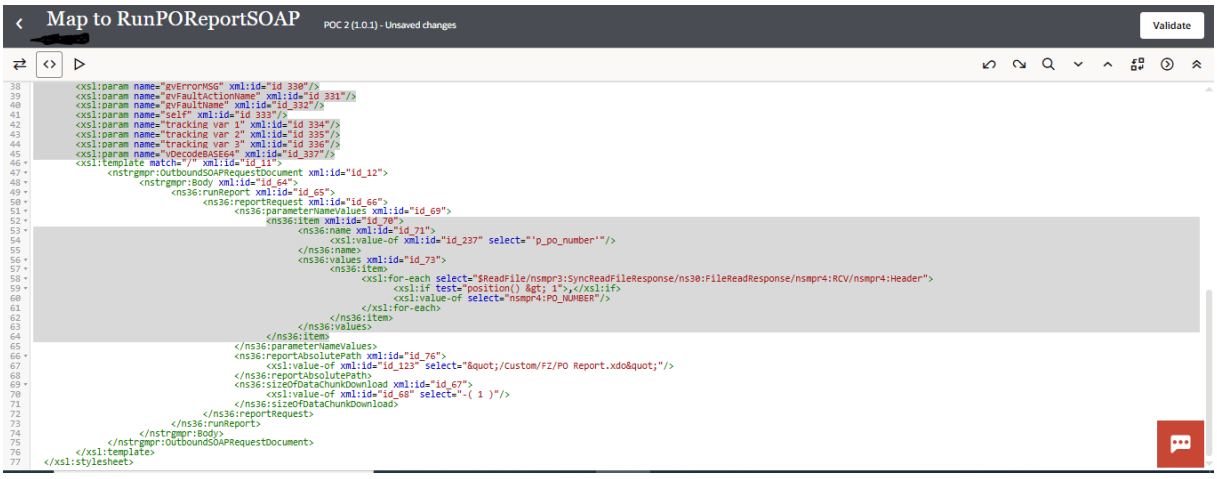
                                    </xsl:for-each>

                                  </ns36:item>

                                </ns36:values>

                              </ns36:item>

##########################################
```



BIP SQL Query (POC2) — REGEXP Solution

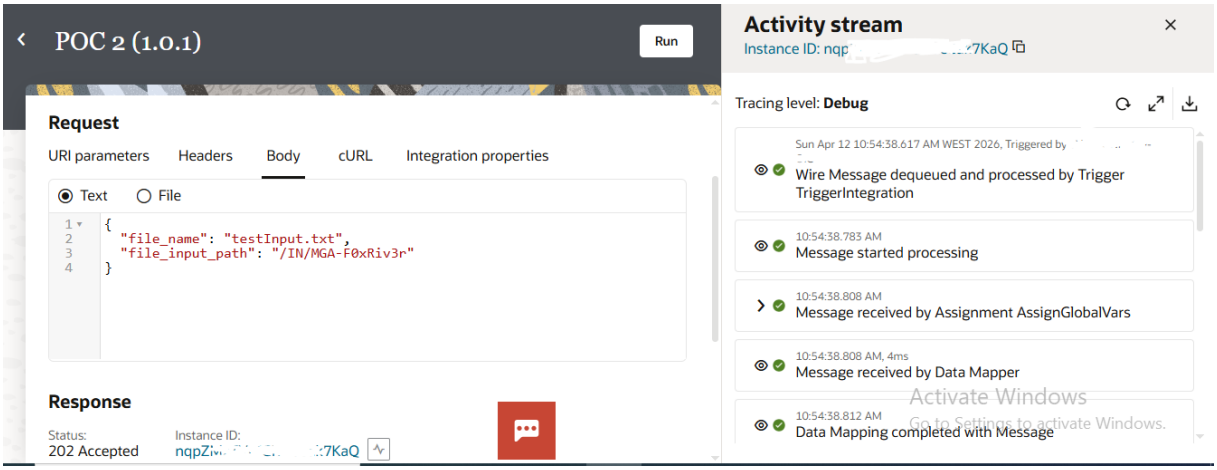
The SQL query is updated to handle a comma-separated list of values :

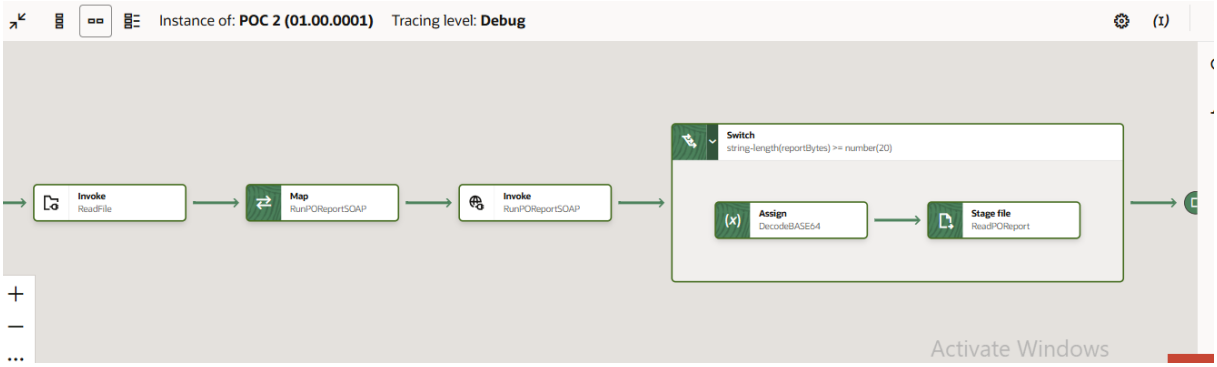
WHERE PO_DATA.SEGMENT1 IN

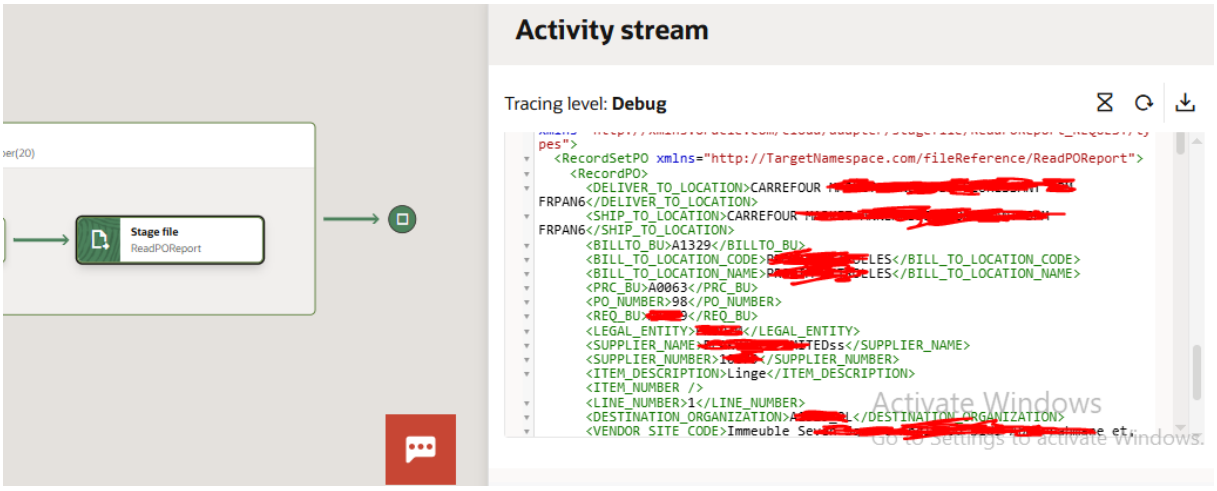
(SELECT REGEXP_SUBSTR(:p_po_number, '[^,]+', 1, LEVEL) FROM DUAL

CONNECT BY REGEXP_SUBSTR(:p_po_number, '[^,]+', 1, LEVEL) IS NOT NULL)

Execution of the POC2 Integration





The Activity Stream confirms that the Map RunPOReportSOAP mapper successfully builds a single SOAP request containing all PO Numbers as a comma-separated string in the p_po_number parameter.

The BIP report response is not empty — the reportBytes field contains data, confirming that Oracle BI Publisher successfully executed the query and returned results for all PO Numbers passed in the single API call.